



Parabola : RACE-01

Single Correct Type Questions:

1. Length of the latus rectum of the parabola $25[(x - 2)^2 + (y - 3)^2] = (3x - 4y + 7)^2$ is-
- (A) 4 (B) 2 (C) $1/5$ (D) $2/5$

Find the vertex, axis, latus rectum, and focus of the parabolas:

MPR208

Subjective Type Questions:

2. Sketch the graph of following parabola Find

- Length of latus rectum
- co-ordinate of focus, vertex,
- ends of latus rectum
- equation of axis, directrix and tangent at vertex.

(i) $y^2 = x$

MPR209

(ii) $y^2 = 10x$

MPR210

(iii) $y^2 = -2x$

MPR211

(iv) $x^2 = 8y$

MPR212

(v) $(y - 1)^2 = 4(x - 2)$

MPR213

(vi) $(x + 3)^2 = 8(1 - 2y)$

MPR214

(vii) $(2y + 1)^2 = 5 - 3x$

MPR215

(viii) $4y^2 + 12x - 20y + 67 = 0$

MPR216

3. Find the equation of following parabola:

(i) focus is (1, 1) and directrix is $y + x + 2 = 0$.

MPR217

(ii) focus is (0, 1) and vertex (0, 2)

MPR218

(iii) length of latus rectum is 8, vertex is (1, 2) and axis is parallel to x-axis

MPR219

(iv) whose latus rectum is 4 units, axis is the line $3x + 4y = 4$ and the tangent at the vertex is the line $4x - 3y + 7 = 0$.

MPR220

(v) Vertex (5, 3) and directrix $3x - 4y + 1 = 0$

MPR221

4. Name the conic represented by the equation $\sqrt{ax} + \sqrt{by} = 1$, where $a, b \in R, a, b, > 0$.

MPR222

5. Prove that the equation to the parabola, whose vertex and focus are on the axis of x at distances a and a' from the origin respectively, is $y^2 = 4(a' - a)(x - a)$.
MPR223
6. PQ is a double ordinate of a parabola $y^2 = 4ax$. Find the locus of its point of trisection.
MPR224
7. A double ordinate of the curve $y^2 = 4px$ is of length $8p$; prove that the lines from the vertex to its two ends are at right angles.
MPR225
8. 'O' is the vertex of the parabola $y^2 = 4ax$ & L is the upper end of the latus rectum. If LH is drawn perpendicular to OL meeting OX in H , prove that the length of the double ordinate through H is $4a\sqrt{5}$.
MPR226
9. A point P on a parabola $y^2 = 4x$, the foot of the perpendicular from it upon the directrix, and the focus are the vertices of an equilateral triangle, find the area of the equilateral triangle.
MPR227
10. Prove that for all values of t , $x = u \cos \alpha t$, $y = u \sin \alpha t - \frac{1}{2}gt^2$ represents a parabola and find its vertex and latus rectum.

ANSWER KEY

1. D
3. (i) $x^2 + y^2 - 2xy - 8x - 8y = 0$ (ii) $x^2 = -4(y - 2)$
- (iii) $(y - 2)^2 = \pm 8(x - 1)^2$ (iv) $\left(\frac{3x + 4y - 4}{5}\right)^2 = \pm 4\frac{(4x - 3y + 7)}{5}$
- (v) $\left(\frac{4x + 3y - 29}{5}\right)^2 = \frac{16}{5}\left(\frac{3x - 4y - 3}{5}\right)$
4. Parabola
6. $9y^2 = 4ax$
9. $4\sqrt{3}$
10. $\left(\frac{u^2 \sin 2\alpha}{2g}, \frac{u^2 \sin^2 \alpha}{2g}\right), LR = \frac{2u^2 \cos^2 \alpha}{g}$



Parabola : RACE-02

Single Correct Type Questions:

- Locus of the feet of the perpendiculars drawn from vertex of the parabola $y^2 = 4ax$ upon all such chords of the parabola which subtend a right angle at the vertex is
 (A) $x^2 + y^2 - 4ax = 0$ (B) $x^2 + y^2 - 2ax = 0$
 (C) $x^2 + y^2 + 2ax = 0$ (D) $x^2 + y^2 + 4ax = 0$ MPR229
- Through the focus of the parabola $y^2 = 2px$ ($p > 0$) a line is drawn which intersects the curve at $A(x_1, y_1)$ and $B(x_2, y_2)$. The ratio $\frac{y_1 y_2}{x_1 x_2}$ equals-
 (A) 2 (B) -1 (C) -4 (D) some function of p MPR230

Subjective Type Questions:

- Find the value of ' a ' for which the point $(a^2 - 1, a)$ lies inside the parabola $y^2 = 8x$. MPR231
- The focal distance of a point on the parabola $(x - 1)^2 = 16(y - 4)$ is 8. Find the co-ordinates. MPR232
- Show that the focal chord of parabola $y^2 = 4ax$ makes an angle α with x-axis is of length $4a \operatorname{cosec}^2 \alpha$. MPR233
- If one end of focal chord of parabola
 (i) $y^2 = 8x$ is $(8, -8)$, find other end. MPR234
 (ii) $y^2 = -2x$ is $(-2, 2)$, find other end. MPR235
 (iii) $x^2 = 4y$ is $(-4, 4)$, find other end. MPR236
 (iv) $(y - 1)^2 = 4 - 8x$ is $(-4, -5)$, find other end. MPR237
- Find the length of focal chord AB of parabola when
 (i) length of latus rectum is 8, ' S ' be the focus and $\frac{AS}{BS} = \frac{3}{4}$ MPR238
 (ii) Length of latus rectum is 4, perpendicular distance of focal chord from vertex is 3. MPR239
 (iii) Length of latus rectum is 2, slope of focal chord is 2 and slope of axis of parabola is -1. MPR240

Parabola

8. Find the condition that the straight line $ax + by + c = 0$ touches the parabola $y^2 = 4kx$. MPR241
9. Find the length of the chord of the parabola $y^2 = 8x$, whose equation is $x + y = 1$. MPR242
10. Prove that the locus of the middle points of all chords of the parabola $y^2 = 4ax$ which are drawn through the vertex is the parabola $y^2 = 2ax$. MPR243

ANSWER KEY

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|---|------------------|---|
| 1. A | 2. C | 3. $\left(-\infty, -\sqrt{\frac{8}{7}}\right) \cup \left(\sqrt{\frac{8}{7}}, \infty\right)$ |
| 4. $(-7, 8), (9, 8)$ | | |
| 6. (i) $(1/2, 2)$ (ii) $(-1/8, -1/2)$ | (iii) $(1, 1/4)$ | (iv) $(-7/8, 11/3)$ |
| 7. (i) $\frac{49}{6}$ (ii) $\frac{4}{9}$ (iii) $\frac{20}{9}$ | 8. $kb^2 = ac$ | 9. $8\sqrt{3}$ |



Parabola : RACE-03

Single Correct Type Questions:

1. The straight line joining any point P on the parabola $y^2 = 4ax$ to the vertex and perpendicular from the focus to the tangent at P , intersect at R , then the equation of the locus of R is
 (A) $x^2 + 2y^2 - ax = 0$ (B) $2x^2 + y^2 - 2ax = 0$
 (C) $2x^2 + 2y^2 - ay = 0$ (D) $2x^2 + y^2 - 2ay = 0$
MPR244
2. Through the vertex O of the parabola, $y^2 = 4ax$ two chords OP & OQ are drawn and the circles on OP & OQ as diameters intersect in R . If θ_1, θ_2 & ϕ are the angles made with the axis by the tangents at P & Q on the parabola & by OR then the value of, $\cot\theta_1 + \cot\theta_2 =$
 (A) $-2 \tan\phi$ (B) $-2\tan(\pi - \phi)$ (C) 0 (D) $2 \cot\phi$
MPR245

Subjective Type Questions:

3. Find the equation of tangent(s) to the parabola
 (i) $y^2 = 2x$ whose slope is $\frac{5}{4}$
MPR246
 (ii) $x^2 = -2y$ which is parallel to $4x + y + 1 = 0$
MPR247
 (iii) $y^2 = 12x$, which passes through the point $(2, 5)$.
MPR248
 (iv) $y^2 = 8x$ and making an angle of 45° with the line $y = 3x + 5$.
MPR249
4. Find the point of contact when
 (i) Equation of parabola is $x^2 = -8y$ and slope of tangent is 3.
MPR250
 (ii) Equation of parabola is $y^2 = 4(1 - x)$ and slope of tangent is -1 .
MPR251
5. Find the point of intersection of tangents drawn at
 (i) the points $(2, -4)$ and $\left(\frac{1}{2}, -2\right)$ of the parabola $y^2 = 8x$
MPR252
 (ii) the points $(-2, 2)$ and $(-8, -4)$ of parabola $y^2 = -2x$.
MPR253
6. Prove that the locus of the point of intersection of tangents to the parabola $y^2 = 4ax$ which meet at an angle θ is $(x + a)^2 \tan^2\theta = y^2 - 4ax$.
MPR254

7. Prove that the straight line $\ell x + my + n = 0$ touches the parabola $y^2 = 4ax$ if $\ell n = am^2$. MPR255
8. Find the equation of common tangent(s) of
(i) Circle $x^2 + y^2 = 4ax$ and parabola $y^2 = 4ax$. MPR256
(ii) Parabola $x^2 = -8y$ and $y^2 = -x$. MPR257
9. Tangents are drawn to the parabola $y^2 = 12x$ at the points A, B and C such that the three tangents form a triangle PQR . If θ_1, θ_2 and θ_3 be the inclinations of these tangents with the axis of x such that their cotangents form an arithmetical progression (in the same order) with common difference 2. Find the area of the triangle PQR . MPR258
10. Two tangents to the parabola $y^2 = 8x$ meet the tangent at its vertex in the points P & Q . If $PQ = 4$ units, prove that the locus of the point of the intersection of the two tangents is $y^2 = 8(x + 2)$. MPR259
11. Let a variable point A be lying on the directrix of parabola $y^2 = 4x$. Tangents AB & AC are drawn to the curve where B & C are points of contact of tangent. If the locus of centroid of $\triangle ABC$ is a conic then find the length of its latus rectum. MPR260
12. Find the locus of the point P from which tangents are drawn to the parabola $y^2 = 4ax$ having slopes m_1 and m_2 such that -
(i) $m_1^2 + m_2^2 = 10$ MPR261
(ii) $\theta_1 - \theta_2 = 30^\circ$ (constant) MPR262
- where θ_1 and θ_2 are the inclinations of the tangents from positive x -axis.

ANSWER KEY

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|--|---|
| 1. B | 2. A |
| 3. (i) $25x - 20y + 8 = 0$ | (ii) $4x + y - 8 = 0$ |
| (iii) $x - y + 3 = 0, 3x - 2y + 4 = 0$ | (iv) $2x + y + 1 = 0, x - 2y + 8 = 0$ |
| 4. (i) $(-12, -18)$ (ii) $(0, 2)$ | 5. (i) $(1, -3)$ (ii) $(4, -1)$ |
| 8. (i) $x = 0$ (ii) $x + 2y - 1 = 0$ | 9. 72 |
| 11. 3 | 12. (i) $10x^2 - y^2 + 2ax = 0$ (ii) $3(y^2 - 4ax) = (x + a)^2$ |



Parabola : RACE-04

Single Correct Type Questions:

1. If three distinct and real normals can be drawn to $y^2 = 8x$ from the point $(a, 0)$, then -
 (A) $a > 2$ (B) $a \in (2, 4)$
 (C) $a > 4$ (D) none of these
MPR263
2. If two normals to a parabola $y^2 = 4ax$ intersect at right angles then the chord joining their feet pass through a fixed point whose co-ordinates are:
 (A) $(-2a, 0)$ (B) $(a, 0)$
 (C) $(2a, 0)$ (D) none
MPR264
3. If the normal to a parabola $y^2 = 4ax$ at P meets the curve again in Q and if PQ and the normal at Q makes angles α and β respectively with the x-axis then $\tan\alpha (\tan\alpha + \tan\beta)$ has the value equal to
 (A) 0 (B) -2
 (C) $-\frac{1}{2}$ (D) -1
MPR265
4. If a normal to a parabola $y^2 = 4ax$ makes an angle ϕ with its axis, then it will cut the curve again at an angle
 (A) $\tan^{-1}(2 \tan\phi)$ (B) $\tan^{-1}\left(\frac{1}{2} \tan\phi\right)$
 (C) $\cot^{-1}\left(\frac{1}{2} \tan\phi\right)$ (D) none
MPR266

Subjective Type Questions:

5. Find the number of distinct normal that can be drawn from $(-2, 1)$ to the parabola $y^2 - 4x - 2y - 3 = 0$.
MPR267
6. If $2x + y + k = 0$ is a normal to the parabola $y^2 = -16x$, then find the value of k .
MPR268
7. Find the equation of normal(s) to the parabola
 (i) $y^2 = -2x + 1$ at the point $(-4, -3)$
MPR269
 (ii) $y^2 = -2x$ and slope of normal is 3.
MPR270
 (iii) $(y - 4)^2 = 16x + 16$ and normal passes through $(14, 7)$
MPR271

Parabola

8. Three normals are drawn from the point $(7, 14)$ to the parabola $x^2 - 8x - 16y = 0$. Find the co-ordinates of the feet of the normals. **MPR272**
9. Write down the equations to the normal at the ends of the latus rectum of the parabola $y^2 = 4a(x - a)$ **MPR273**
10. Find the point of the parabola $y^2 = 4ax$ at which the normal is inclined at 30° to the axis. **MPR274**

ANSWER KEY

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|------------------------------------|-------------------------------|---|
| 1. C | 2. B | 3. B |
| 4. B | 5. 1 | 6. 48 |
| 7. (i) $3x + y + 15 = 0$ | (ii) $6x - 2y + 33 = 0$ | (iii) $x = 2y, 3x + 2y = 56, x - y - 7 = 0$ |
| 8. $(0, 0), (-4, 3)$ and $(16, 8)$ | 9. $(x + y = 4a, x - y = 4a)$ | 10. $\left(\frac{a}{3}, -\frac{2\sqrt{3}}{3}a\right)$ |



Parabola : RACE-05

Single Correct Type Questions:

- Normal to the parabola $y^2 = 8x$ at the point $P(2, 4)$ meets the parabola again at the point Q . If C is the centre of the circle described on PQ as diameter then the coordinates of the image of the point C in the line $y = x$ are
 (A) $(-4, 10)$ (B) $(-3, 8)$
 (C) $(4, -10)$ (D) $(-3, 10)$ MPR275
- PQ is a normal chord of the parabola $y^2 = 4ax$ at P , A being the vertex of the parabola. Through P a line is drawn parallel to AQ meeting the x -axis in R . Then the length of AR is -
 (A) equal to the length of the latus rectum.
 (B) equal to the focal distance of the point P .
 (C) equal to twice the focal distance of the point P .
 (D) equal to the distance of the point P from the directrix. MPR276
- Normals are drawn at points A , B , and C on the parabola $y^2 = 4x$ which intersect at $P(h, k)$. The locus of the point P if the slope of the line joining the feet of two of them is 2, is
 (A) $x + y = 1$ (B) $x - y = 3$
 (C) $y^2 = 2(x - 1)$ (D) $y^2 = 2\left(x - \frac{1}{2}\right)$ MPR277
- Tangents are drawn from the point $(-1, 2)$ on the parabola $y^2 = 4x$. The length, these tangents will intercept on the line $x = 2$ is :
 (A) 6 (B) $6\sqrt{2}$ (C) $2\sqrt{6}$ (D) none of these MPR278

Subjective Type Questions:

- Three normals to $y^2 = 4x$ pass through the point $(15, 12)$. Show that if one of the normals is given by $y = x - 3$ & find the equations of the others. MPR279
- Prove that, the normal to $y^2 = 12x$ at $(3, 6)$ meets the parabola again in $(27, -18)$ & find the equation of circle on this normal chord as diameter. MPR280
- From the point $P(h, k)$ three normals are drawn to the parabola $x^2 = 8y$ and m_1, m_2 and m_3 are the slopes of three normals
 (a) Find the algebraic sum of the slopes of these three normals. MPR281
 (b) If two of the three normals are at right angles then the locus of point P is a conic, find the latus rectum of conic. MPR282
 (c) If the two normals from P are such that they make complementary angles with the axis then the locus of point P is a conic, find a directrix of conic. MPR283

Parabola

8. PC is the normal at P to the parabola $y^2 = 4ax$, C being on the axis. CP is produced outwards to Q so that $PQ = CP$; show that the locus of Q is a parabola. MPR284
9. Let $P(a, b)$ and $Q(c, d)$ are the two points on the parabola $y^2 = 8x$ such that the normals at them meet in $(18, 12)$. Find the product $(abcd)$. MPR285
10. Find combined equation of pair of tangents drawn to parabola.
(i) $y^2 = 8x$ from $(-3, 1)$ MPR286
(ii) $(x - 1)^2 = 4(y - 2)$ from $(2, 0)$ MPR287
11. If the equation $\lambda^2x + \lambda y - \lambda^2 + 2\lambda + 7 = 0$ represents a family of lines, where ' λ ' is parameter, then find the equation of the curve to which these lines will always be tangents. MPR288
12. Find the angle between the tangents drawn from the origin to the parabola, $y^2 = 4a(x - a)$. MPR289

ANSWER KEY

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|--|--|--|---------------|
| 1. A | 2. C | 3. B | 4. B |
| 5. $y = -4x + 72, y = 3x - 33$ | 6. $x^2 + y^2 - 30x + 12y - 27 = 0$ | 7. (a) 0; (b) 2; (c) $y = \frac{3}{2}$ | 9. 512 |
| 10. (i) $[2x^2 - 3y^2 - xy + 13x + 3y + 18 = 0]$ | (ii) $[2x^2 - y^2 + xy - 8x - 2y + 8 = 0]$ | 11. $(y + 2)^2 = 28(x - 1)$ | 12. $(\pi/2)$ |



Parabola : RACE-06

Single Correct Questions

1. TP & TQ are tangents to the parabola, $y^2 = 4ax$ at P & Q . If the chord PQ passes through the fixed point $(-a, b)$ then the locus of T is :

(A) $ay = 2b(x - b)$

(B) $bx = 2a(y - a)$

(C) $by = 2a(x - a)$

(D) $ax = 2b(y - b)$

MPR293

Subjective Type Questions:

2. Find the equation of the chord of contacts of tangents drawn from a point $(2, 1)$ to the parabola $x^2 = 2y$.

MPR290

3. Find the co-ordinates of the middle point of the chord of the parabola $y^2 = 16x$, the equation of which is $2x - 3y + 8 = 0$

MPR291

4. Prove that the length of the chord joining the points of contact of tangents drawn from the point

$$(x_1, y_1) \text{ is } \frac{\sqrt{y_1^2 + 4a^2} \sqrt{y_1^2 - 4ax_1}}{a}$$

MPR292

5. Through the vertex O of a parabola $y^2 = 4x$, chords OP & OQ are drawn at right angles to one another. Show that for all positions of P , PQ cuts the axis of the parabola at a fixed point. Also find the locus of the middle point of PQ .

MPR294

6. From the point $(-1, 2)$ tangent lines are drawn to the parabola $y^2 = 4x$. Find the equation of the chord of contact. Also find the area of the triangle formed by the chord of contact & the tangents.

MPR295

7. PQ , a variable chord of the parabola $y^2 = 4x$ subtends a right angle at the vertex. The tangents at P and Q meet at T and the normals at those points meet at N . If the locus of the mid point of TN is a parabola, then find its latus rectum.

MPR296

8. Prove that the normal chord at the point whose ordinate is equal to its abscissa subtends a right angle at the focus.

MPR297

9. Prove that the locus of the middle points of all tangents drawn from points on the directrix to the parabola is $y^2(2x + a) = a(3x + a)^2$.

MPR298

- 10.** A variable circle passes through the point $A(2,1)$ and touches the x-axis. Locus of the other end of the diameter through A is a parabola.
 (a) Find the length of the latus rectum of the parabola. **MPR299**
 (b) Find the co-ordinates of the foot of the directrix of the parabola. **MPR300**
 (c) The two tangents and two normals at the extremities of the latus rectum of the parabola constitutes a quadrilateral. Find area of quadrilateral. **MPR301**
- 11.** Let $L_1 : x + y = 0$ and $L_2 : x - y = 0$ are tangent to a parabola whose focus is $S(1, 2)$. If the length of latus-rectum of the parabola can be expressed as $\frac{m}{\sqrt{n}}$ (where m and n are coprime) then find the value of $(m + n)$. **MPR302**
- 12.** Tangent drawn at point $P(1,3)$ of a parabola intersects its tangent at vertex at $A(-1,5)$. If $B(-5,5)$ is a point on SP where S is focus of the parabola then find
 (i) focus **MPR303**
 (ii) equation of directrix. **MPR304**
 (iii) co-ordinate of vertex. **MPR305**

ANSWER KEY

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|--|--------------------------|------------------|--------------------------|
| 1. C | 2. $2x = y + 1$ | 3. 14, 12 | 5. $y^2 = 2x - 8$ |
| 6. $y = x - 1, 8\sqrt{2}$ | 7. $\frac{25}{2}$ | | |
| 10. (a) 4, (b) $(2, -1)$, (c) 8 sq. units | 11. 11 | | |
| 12. (i) $(-2, 4)$ (ii) $x - 3y + 18 = 0$ (iii) $\left(-\frac{11}{5}, \frac{23}{5}\right)$ | | | |